

STAT 240

Week 13: Review

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Today

- Working as a data scientist
- Future courses you might enjoy
- Review for the written exam
- That's it!

Coding Exam

- Will post the cover page (with data, packages) in Canvas announcement a few minutes before the exam goes live on crowdmark
- Make sure you have everything set up, come and take the exam in the room if unsure
 - I can't help you remotely

Course Experience Survey

- Please fill it out
- 5 minutes to do it now if you haven't already done it
- I want to make this course as useful as possible, want feedback on it

Working as a Data Scientist

- Data science jobs can be very varied, depending on the company
- Can be a glorified “business analyst” up to doing advanced ML tasks
- This is something to keep in mind when you look for Data Science Jobs
 - If the person writing the job ad doesn’t know what data science is, you probably don’t want to work for them

Data Science Careers - Key Skills

- Solid understanding of standard statistical methods
 - Seen a few in this course, 452 more thorough
- Be able to implement these (possibly Python, maybe R)
- Be able to debug code!
- Be able to understand data, and the subtleties that come with real data!
- This is one way to have real impact as a data scientist

More Key Skills

- Being able to communicate
 - With the people who collect/maintain the data
 - With the people who want output from you using the data
 - These are not easy tasks
- The career progression as a data scientist requires being able to understand methods, manage/explain stats and data and even more communicating

Data Science in the Future

- AI tools can solve some of these types of problems, but long way from being able to do all of them well
- Still many interesting/challenging roles in data science
- You need to be able to learn new things regularly as a data scientist, this will only continue
- AI is making this skill even more valuable
 - It will likely reduce the number of data science roles, so you need to be able to distinguish yourself

Future Courses you could/should take

Stat 440 - Learning from Big Data

- A hands on course working with data and modern Data Science tools
- Get to work on projects, which you can use for demonstrating your ability
- Stat 350 and Stat 260 prereqs, 240 also recommended

Stat 452 - Statistical Learning and Prediction

- Much more detailed coverage of prediction, advanced models for prediction
- Cover models like KNN, k-means and other more modern tools
- All the statistical methods you need for ~75% of data science problems
- 260 and 302/305 prereqs

Stat 360 - Advanced R for Data Science

- More details on advanced programming in R
- Version control, data structures in R
- Writing better/faster code, writing R functions/packages
- 260 and 302/305 prereqs

Other Courses

- Stat 445 - More details on clustering
- Stat 475 - Models for dealing with binary/multiclass data
- Stat 485 - Dealing with Time Series Data

Brief Review

Basics

- File formats, reading files
- Essentials of R programming
- Basic statistical functions (mean, min, max, sd, median)

Markdown

- How does a QMD file work
 - Accessing data?
 - Running code?
 - Reproducible?
- Working directories
- How to not run code using `#| eval`, `#| echo`, etc
 - Slide 25 in Week 4

Relational Databases

- Why relational databases?
- What are examples of a relational database?
- How a company might use a relational database?

Linear Regression

- Basic optimization
- What does linear regression try to minimize/maximize
- Checking a regression model

KNN Regression and Prediction

- Measuring prediction error
- KNN
 - How to get estimates for new x
 - How these change depending on K
- Why we split into training/test data

APIs

- Why use APIs?
- How do APIs extract data?
- What do you need to use an API?

Cross- Validation

- Why do we do cross-validation?
- How does CV work for getting overall predictions on test data?
- Know the ins and outs of this

Classification

- Difference with regression models
- How do we measure performance of classification?
- KNN for classification

Clustering

- What is clustering?
- What is unsupervised learning?
- Know the steps in the k-means algorithm
- How to stop running k-means
- Running k-means from multiple random starting points

Strings

- Why is string manipulation useful?
- Understand the ideas behind strings, won't ask you to write regular expressions

Likelihood and Mixture Models

- Understand what the likelihood is, why we maximise it
- What are the parameters in the model
- Is the log likelihood equivalent?
- Why are mixture models more complicated?
- Why do we run the EM algorithm multiple times?

HTML

- What is HTML and why is it useful to know?

Types of Problems

Problem Format

- Some True/False questions, some short answer questions
- Some slightly longer questions
- Keep your answers concise and exact, don't need to write an essay
- Bullet point answers ok